Exercise Sheet

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Throughout, $T \in \mathcal{L}(V)$ and $\chi_T(\lambda) = \det(\lambda I - T)$ is its characteristic polynomial. Tags indicate the flavour of each problem; a ⋆ marks a harder one.

Core tutorial route

Work **6.1(i)–(ii), 6.2(i), 6.3, 6.4(i), 6.5, 6.7, 6.10(ii)–(iii), and 6.10(iv)** in that order. This route covers diagonalization and matrix powers, a common eigenbasis, the Cayley–Hamilton theorem and inverses, spectral mapping, and linear dynamics from an initial state.

Additional practice: 6.1(iii)–(v), 6.2(ii), 6.4(ii)–(iii), 6.6, 6.8(i)–(ii), 6.10(i). **Bridge:** 6.9(ii) (convergence). **Extension:** 6.8(iii), 6.9(i), 6.11–6.13 (the determinant–trace identity, projections, generalized eigenvectors, the Jordan–Chevalley decomposition, the exponential of a defective matrix, ladder operators).

Diagonalization and commuting operators

Exercise 6.1 (computational) Decide whether the matrix A is diagonalizable and, if it is, find matrices P and D such that $P^{-1}AP = D$, where (i) $A = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 2 \end{smallmatrix}\right)$; (ii) $A = \left(\begin{smallmatrix} 2 & 2 \\ 2 & -1 \end{smallmatrix}\right)$ (over $\mathbb{R}$); (iii) $A = \left(\begin{smallmatrix} 2 & -1 \\ 1 & 2 \end{smallmatrix}\right)$ (over $\mathbb{C}$); (iv) $A = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 1 \end{smallmatrix}\right)$; (v) $A = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{smallmatrix}\right)$.

Answer. (i) Eigenvalues $1, 4$; $P = \left(\begin{smallmatrix} 2 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $D = \operatorname{diag}(4, 1)$. (ii) Eigenvalues $3, -2$; $P = \left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$, $D = \operatorname{diag}(3, -2)$. (iii) Eigenvalues $2 \pm i$; $P = \left(\begin{smallmatrix} 1 & 1 \\ -i & i \end{smallmatrix}\right)$, $D = \operatorname{diag}(2+i, 2-i)$. (iv) Not diagonalizable: the only eigenvalue 2 has a one-dimensional eigenspace. (v) Not diagonalizable: the only eigenvalue 2 has a one-dimensional eigenspace.

Exercise 6.2 (computational) Compute A^n for every $n \geq 0$, where (i) $A = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 2 \end{smallmatrix}\right)$; (ii) $A = \left(\begin{smallmatrix} 2 & 2 \\ 2 & -1 \end{smallmatrix}\right)$.

Answer. (i) $A^n = \frac{1}{3}\left(\begin{smallmatrix} 2\cdot 4^n + 1 & 2\cdot 4^n - 2 \\ 4^n - 1 & 4^n + 2 \end{smallmatrix}\right)$. (ii) $A^n = \frac{1}{5}\left(\begin{smallmatrix} 4\cdot 3^n + (-2)^n & 2\cdot 3^n - 2(-2)^n \\ 2\cdot 3^n - 2(-2)^n & 3^n + 4(-2)^n \end{smallmatrix}\right)$.

Exercise 6.3 (computational) Let $S = \operatorname{diag}(1, 1, 2)$ and $T = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right)$. (i) Check that $ST = TS$. (ii) The standard basis consists of eigenvectors of S. Does it consist of eigenvectors of T? (iii) Find a basis of $\mathbb{R}^3$ in which the matrices of both S and T are diagonal.

Answer. (i) $ST = TS = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 8 \end{smallmatrix}\right)$. (ii) No: $Te_1 = (2, 1, 0)$. (iii) $(1, 1, 0)$, $(1, -1, 0)$, $(0, 0, 1)$; in it $S = \operatorname{diag}(1, 1, 2)$ and $T = \operatorname{diag}(3, 1, 4)$.

Hint. (iii) The operator T maps the eigenspace $E(1, S) = \operatorname{span}(e_1, e_2)$ into itself. Look for the eigenvectors of T inside it.

The Cayley–Hamilton theorem

Exercise 6.4 (computational) Use the Cayley–Hamilton theorem to express the matrix A^{-1} as a polynomial in A, and compute it, where (i) $A = \left(\begin{smallmatrix} 2 & 3 \\ 4 & 1 \end{smallmatrix}\right)$; (ii) $A = \left(\begin{smallmatrix} 3 & 1 \\ 2 & 4 \end{smallmatrix}\right)$; (iii) $A = \left(\begin{smallmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 3 \end{smallmatrix}\right)$ (compute the characteristic polynomial χ_A first).

Answer. (i) $A^{-1} = \frac{1}{10}(A - 3I) = \begin{pmatrix} -\frac{1}{10} & \frac{3}{10} \\ \frac{2}{5} & -\frac{1}{5} \end{pmatrix}$. (ii) $A^{-1} = \frac{1}{10}(7I - A) = \begin{pmatrix} \frac{2}{5} & -\frac{1}{10} \\ -\frac{1}{5} & \frac{3}{10} \end{pmatrix}$. (iii) $\chi_A(\lambda) = \lambda^3 - 8\lambda^2 + 21\lambda - 18$; $A^{-1} = \frac{1}{18}(A^2 - 8A + 21I) = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ -\frac{1}{6} & \frac{1}{2} & 0 \\ \frac{1}{18} & -\frac{1}{6} & \frac{1}{3} \end{pmatrix}$.

Exercise 6.5 (computational) Let $A = \left(\begin{smallmatrix} 0 & 1 \\ -2 & 3 \end{smallmatrix}\right)$. Use the Cayley–Hamilton theorem to write A^4 in the form $\alpha A + \beta I$, and compute the resulting matrix.

Answer. $A^4 = 15A - 14I = \left(\begin{smallmatrix} -14 & 15 \\ -30 & 31 \end{smallmatrix}\right)$.

Exercise 6.6 (proof) Let $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Show by direct computation that $A^2 - (\operatorname{tr} A)\,A + (\det A)\,I = 0$. This verifies the Cayley–Hamilton theorem for every 2×2 matrix.

Hint. Compute A^2 and $(\operatorname{tr} A)A$ and subtract; the off-diagonal entries cancel, and what is left is a multiple of I.

Functions of operators and linear dynamics

Exercise 6.7 (computational) Let $A = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 2 \end{smallmatrix}\right)$, with $\operatorname{spec}(A) = \{1, 4\}$. Use spectral mapping to find $\operatorname{spec}(B)$ for $B = A^3 - 2A + I$.

Answer. $\operatorname{spec}(B) = \{0, 57\}$ (with the same eigenvectors $(1, -1)$ and $(2, 1)$).

Exercise 6.8 (proof)

(i) Suppose $T \in \mathcal{L}(V)$ is diagonalizable and has only one eigenvalue λ. Prove that $T = \lambda I$.

(ii) Suppose T is diagonalizable and $T = PDP^{-1}$. Prove that for any polynomials p, q the operators $p(T)$ and $q(T)$ commute.

(iii) Let A be a diagonalizable matrix. Prove that $\det(e^A) = e^{\operatorname{tr} A}$.

Hint. (i) Diagonalize: $T = PDP^{-1}$, where every diagonal entry of D is the same λ, and see what that leaves. (ii) Move everything to D: $p(T) = P\,p(D)\,P^{-1}$, and diagonal matrices commute. Note also that diagonalizability is not actually needed — it is enough that the powers of T commute. (iii) A similarity transformation changes neither the determinant nor the trace, and $e^{PDP^{-1}} = P\,e^D P^{-1}$, where e^D is a diagonal matrix. What remains is $\prod_k e^{\lambda_k}$.

Exercise 6.9 ($\star$)

(i) Suppose T is diagonalizable and its eigenvalues lie in the set $\{0, 1\}$. Prove that $T^2 = T$.

(ii) Suppose A is a diagonalizable matrix and each of its eigenvalues satisfies $|\lambda| < 1$. Prove that $A^n \to 0$ as $n \to \infty$ (componentwise).

Hint. (i) Diagonalize T; the claim reduces to checking $\lambda^2 = \lambda$ for each eigenvalue. (ii) Diagonalize and examine D^n componentwise; the matrices P and P^{-1} do not depend on n.

Exercise 6.10 (computational)

(i) Compute e^{tA}, where $A = \left(\begin{smallmatrix} 3 & 2 \\ 1 & 2 \end{smallmatrix}\right)$.

(ii) Compute e^{tA}, where $A = \left(\begin{smallmatrix} 2 & 2 \\ 2 & -1 \end{smallmatrix}\right)$.

(iii) Solve the equation $\dot{x} = Ax$ with $A = \left(\begin{smallmatrix} 2 & 2 \\ 2 & -1 \end{smallmatrix}\right)$ and $x(0) = (1, 3)$.

(iv) Let $A = \left(\begin{smallmatrix} 0 & -4 \\ 1 & 0 \end{smallmatrix}\right)$. Compute e^{tA} and describe the trajectories of the solutions of $\dot{x} = Ax$.

Answer. (i) $e^{tA} = \frac{1}{3}\begin{pmatrix} 2e^{4t}+e^{t} & 2e^{4t}-2e^{t} \\ e^{4t}-e^{t} & e^{4t}+2e^{t} \end{pmatrix}$. (ii) $e^{tA} = \frac{1}{5}\begin{pmatrix} 4e^{3t}+e^{-2t} & 2e^{3t}-2e^{-2t} \\ 2e^{3t}-2e^{-2t} & e^{3t}+4e^{-2t} \end{pmatrix}$. (iii) $x(t) = \big(2e^{3t} - e^{-2t},\ e^{3t}+2e^{-2t}\big)$. (iv) $e^{tA} = \begin{pmatrix} \cos 2t & -2\sin 2t \\ \frac{1}{2}\sin 2t & \cos 2t \end{pmatrix}$; the trajectories other than the stationary origin are ellipses $x_1^2 + 4x_2^2 = C$, traversed counterclockwise at angular frequency 2.

Generalized eigenvectors *[self-study]*

Exercise 6.11 (computational) Find the eigenspace $E(2, A)$ and the generalized eigenspace $G(2, A)$ of the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ of Exercise 6.1(v). Write down the Jordan chain.

Answer. $E(2, A) = \operatorname{span}\big((1, 0, 0)\big)$ $(\dim 1)$, $G(2, A) = \mathbb{C}^3$ $(\dim 3)$; Jordan chain $(0, 0, 1) \to (0, 1, 0) \to (1, 0, 0) \to 0$.

Exercise 6.12 (computational)

(i) Write the matrix $A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$ of Exercise 6.1(iv) in the form $A = D + N$, where D is diagonalizable, N is nilpotent and $DN = ND$. Use this decomposition to compute e^{tA}.

(ii) Compute e^{tA} for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Answer. (i) $D = 2I$, $N = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$; $e^{tA} = e^{2t}\begin{pmatrix} 1+t & t \\ -t & 1-t \end{pmatrix}$. (ii) $e^{tA} = \begin{pmatrix} e^{2t} & te^{2t} & 0 \\ 0 & e^{2t} & 0 \\ 0 & 0 & e^{3t} \end{pmatrix}$.

Exercise 6.13 ($\star$) On the state space of spin 1, the raising operator acts by $J_+|{-1}\rangle = \sqrt{2}\,|0\rangle$, $J_+|0\rangle = \sqrt{2}\,|1\rangle$, $J_+|1\rangle = 0$. Write its matrix in the basis $(|1\rangle, |0\rangle, |{-1}\rangle)$ and show that J_+ is nilpotent.

Hint. The columns of the matrix are the images of the basis kets. A strictly upper triangular matrix is nilpotent — compute its powers.